

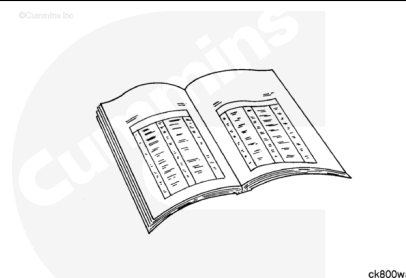
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the vehicle driveline and transmission. Refer to the OEM service manual.
- Remove the clutch discs and pressure plate. Refer to the OEM service manual.

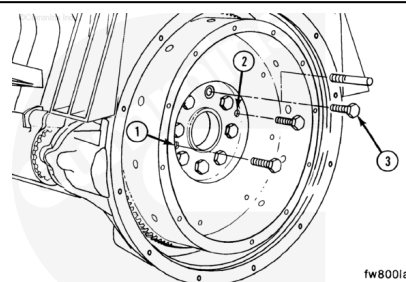


Remove

Hold the flywheel to prevent rotation.

Install M10 - 1.50 x 40 puller capscrews (at points 1 and 2).

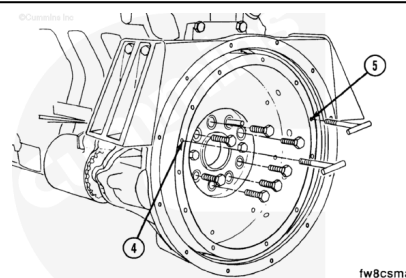
Remove capscrew (3) and install guide pin, Part Number 3376696.



Note : If a clutch is used in the equipment, the threads in the clutch pressure plate mounting capscrew holes can be metric or standard. Be sure to use the correct capscrews.

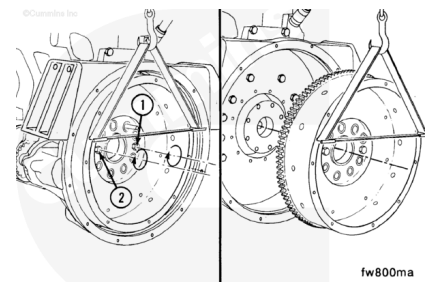
Determine the capscrew thread design and size, and install two "t-handles" in the flywheel (at points 4 and 5).

Remove the remaining seven flywheel mounting capscrews.



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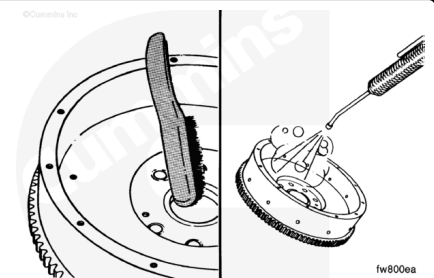
Tighten capscrews (1 and 2) in alternating sequence to loosen the flywheel.

Remove the flywheel from the guide pin.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

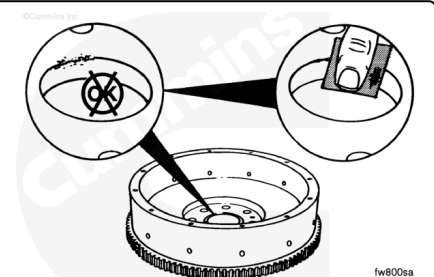
When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Use a wire brush to clean the crankshaft pilot bore.

Use steam or solvent to clean the flywheel. Dry with compressed air.

Inspect for nicks or burrs.

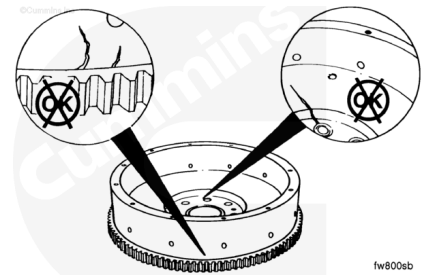
Use Scotch-Brite™ 7448, Part Number 3823258 or equivalent, to remove small nicks and burrs.



⚠ WARNING ⚠

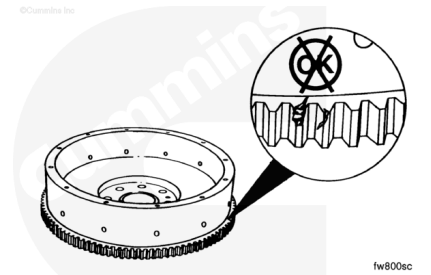
Do not use a cracked flywheel. A cracked flywheel can break and cause serious personal injury or property damage.

Inspect the flywheel for cracks.



Inspect the flywheel ring gear teeth for cracks and chips.

If the ring gear teeth are cracked or broken, the ring gear **must** be replaced.

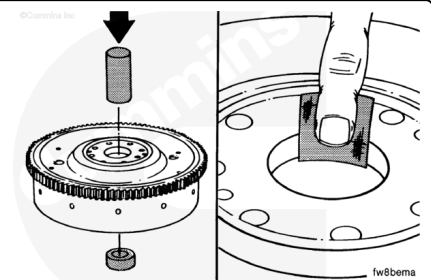


Install

Use a new pilot bearing when installing a new or rebuilt clutch.

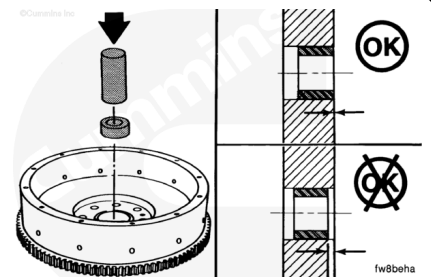
Remove the pilot bearing with a mandrel and hammer.

Clean the pilot bore with Scotch-Brite™ 7448, Part Number 3823258, or equivalent.



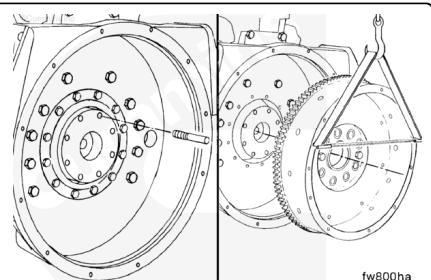
Install the pilot bearing with a mandrel and hammer.

The pilot bearing **must** be installed flush with the pilot bore surface.



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Install guide pin, Part Number 3376696, in the crankshaft flange.

Install the flywheel on the guide pin.

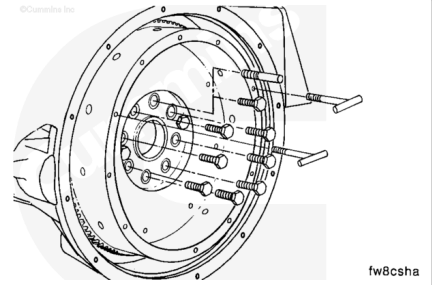
Install the seven capscrews.

Remove the "T-handles" and the guide pin.

Install the remaining capscrew in the hole.

Tighten the capscrews in a star pattern.

Torque Value: 183 n•m [135 ft-lb]



Measure

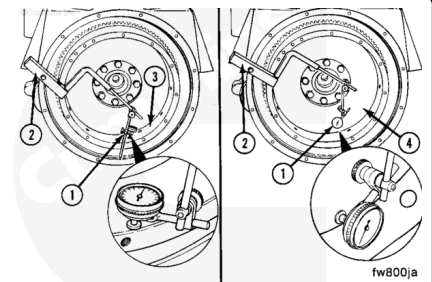
Bore Runout

Use dial indicator gauge (1), Part Number 3376050 or equivalent, and dial gauge attachment (2), Part Number ST-1325, to inspect the flywheel bore (3) and face (4) runout.

Install the attachment to the flywheel housing.

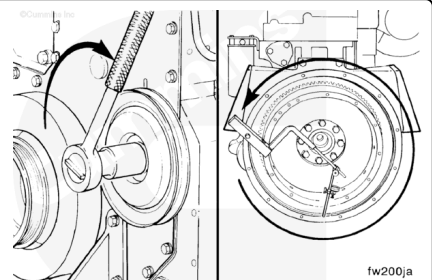
Install the gauge on the attachment.

Install the contact tip of the indicator against the inside diameter of the flywheel bore.



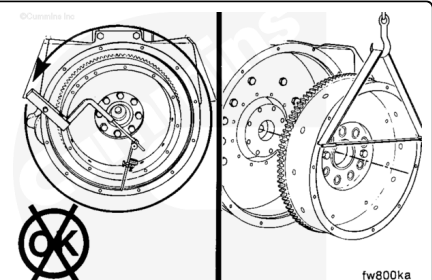
Use the accessory driveshaft to rotate the crankshaft one complete revolution (360 degrees).

The total indicator reading **must not** exceed 0.127 mm [0.0050 in].



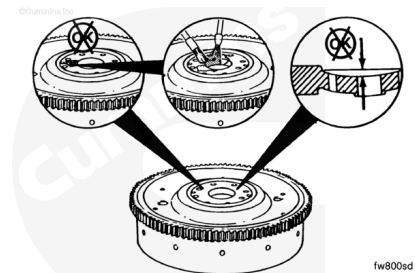
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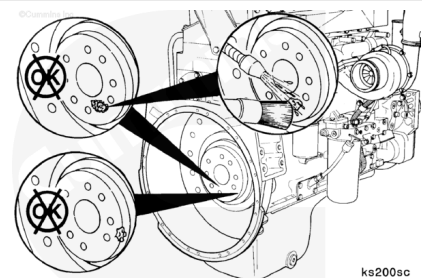


Remove the flywheel if the total indicator reading is greater than the specification.

Inspect the flywheel mounting surface for dirt or damage.

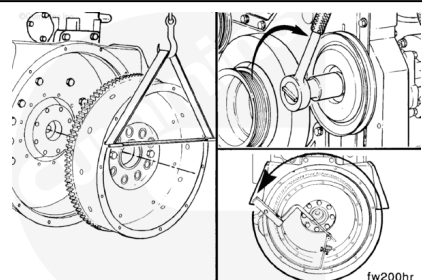


Inspect the crankshaft for dirt or damage.



⚠ WARNING ⚠

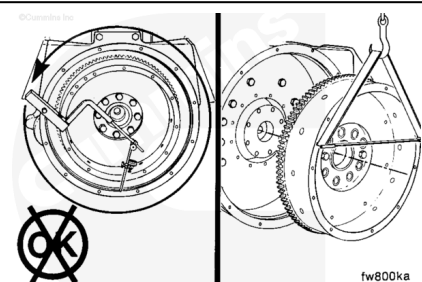
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Install the flywheel and inspect the bore runout again.

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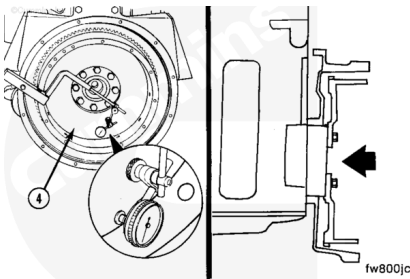


Replace the flywheel if the runout does **not** meet specifications.

Face Runout

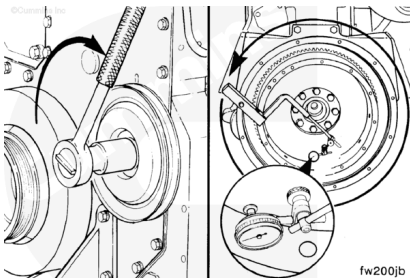
Install the contact tip of the indicator against the flywheel face as close to the outside diameter as possible to inspect the face (4) run out.

Push the flywheel forward to remove the crankshaft end thrust.



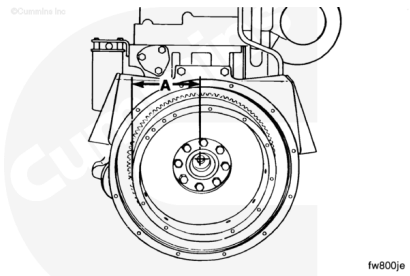
Use the accessory driveshaft to rotate the crankshaft one complete revolution (360 degrees) while keeping the crankshaft end thrust pushed forward.

The total indicator reading **must not** exceed the following specifications.



Flywheel Face Runout Total Indicator Reading

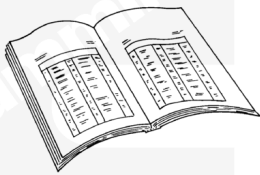
Flywheel Radius (A)		Maximum Total Indicator Reading of Flywheel Face	
mm	in	mm	in
203	8	0.203	0.008
254	10	0.254	0.010
305	12	0.305	0.012
356	14	0.356	0.014
406	16	0.406	0.016



Finishing Steps

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- Install the clutch discs, pressure plate, transmission, and driveline, if equipped, in reverse order of removal. Refer to the OEM service manual.
- Align the universal joints on each end of the driveshaft to prevent vibration.

- Operate the engine and check for proper operation.

Last Modified: 07-May-2009
